Jammming Project
V1 Design Document

Muzaffer Karakuş

Table of Contents

1. Functionalities 5
2. Function Components 7
3. Events 8
4. Detailed Design 10
   4.1. App 11
      4.1.1. Functionalities 11
      4.1.2. JSX 11
   4.2. Login 12
      4.2.1. Functionalities 12
      4.2.2. JSX 12
   4.3. After Login 13
      4.3.1. Functionalities 13
      4.3.2. JSX 13
   4.4. Player 14
      4.4.1. Functionalities 14
      4.4.2. JSX 15
   4.5. Search 18
      4.5.1. Functionalities 18
      4.5.2. JSX 18
   4.6. Results 19
      4.6.1. Functionalities 19
      4.6.2. JSX 19
   4.7. Playlists 21
      4.7.1. Functionalities 21
      4.7.2. JSX 21
   4.8. Tracklist 24
      4.8.1. Functionalities 24
      4.8.2. JSX 24
   4.9. States 27
   4.10. Helper Functions 27
      4.10.1. Login 27
      4.10.2. AfterLogin 27
      4.10.3. Player 27
      4.10.4. PlayerDevices 27
      4.10.5. PlayerDevicesElement 27
      4.10.6. PlayerVolume 27
      4.10.7. PlayerProgressBar 27
      4.10.8. Search 27
      4.10.9. ResultsSong 27
      4.10.10. Playlists 27
      4.10.11. PlaylistsElement 27
      4.10.12. PlaylistsCreate 27
      4.10.13. Tracklist 27
      4.10.14. TracklistTrack 27
      4.10.15. TracklistModify 27
   4.11. Effects Section 27
      4.11.1. Login 27

4.11.2. AfterLogin 27
4.11.3. Player 27
4.11.4. PlayerDevices 27
4.11.5. PlayerVolume 27
4.11.6. PlayerProgressBar 27
4.11.7. Results 27
4.11.8. ResultsSong 27
4.11.9. Playlists 27
4.11.10. PlaylistsElement 27
4.11.11. Tracklist 27
4.11.12. TracklistTrack 27
4.11.13. TracklistModify 27
5. Seperation of Presentational and Container Components 28
6. Design of CSS 28

List of Tables

Table 1 Events in **Login Component and its Children** 8
Table 2 Events in **Search Component and its Children** 8
Table 3 Events in **Results Component and its Children** 8
Table 4 Events in **Playlist Component and its Children** 8
Table 5 Events in **Tracklist Component and its Children** 8
Table 6 Events in **Player Component and its Children** 9
Table 7 States table 27

List of Figures

Figure 1 Component Chart 7
Figure 2 App jsx 11
Figure 3 Login jsx 12
Figure 4 AfterLogin jsx 13
Figure 5 Player jsx 15
Figure 6 Search jsx 18
Figure 7 Results jsx 19
Figure 8 Playlists jsx 21
Figure 9 Tracklist jsx 24
Figure 10 Helper functions 27
Figure 11 Player Devices 30
Figure 12 Results Main Element 31
Figure 13 Rename Playlist Element 32
Figure 14 Create Playlist Element 33
Figure 15 Tracklist main element 34
Figure 16 Modify Tracklist Element 35

1. Functionalities

Think about your apps general functionalities and write them down below. Write down the http request that will be required as a sub element to each functionality. Research through required login steps, and all the http request that will be required. Link the related documents beside each http request. When you update your whole app(When adding new functionalities, removing or updating functionalities), add new functionalities and related http requests at the bottom of the list following the same format.

- Login
  - Token Request
- Searching for a track
  - Search Request
- Displaying search results
- Adding a track to the playlist
  - Add a Track Request
- Removing a track from the playlist
  - Remove a Track Request
- Displaying your playlists
  - Playlists Request
- Creating a playlist
  - Create a Playlist Request
- Renaming playlist
  - Rename a Playlist Request
- Displaying playlist tracks
  - Playlist Elements Request
- Using a player
  - Player Request
- Playing a specific song from search results
  - Play a spesific song request
- Playing a song from playlist
  - Play a spesific song from playlist
- Playing a playlist
  - Play a playlist
- Changing volume
  - Volume Change Request
- Getting Current Volume
  - Getting Volume Request
- Displaying current song played, current playlist played, previous song, and next song
  - Current state request
- Changing current song position using progress bar
  - Seeking a position request

- Skipping in to previous
  - Skip to Previous Request
- Skipping to next song on a playlist
  - Skip to Next Request
- Playing in mixed order in a playlist
  - Shuffle Request
- Playing a song repetitively
  - Repeat song request
- Playing a list repetitively
  - Repeat playlist request
- Displaying current device and active devices
  - Get Active Devices Request
- Selecting current device
  - Change current device request
- Pausing a song
  - Pause request
- Resuming a song
  - Resume request
- Displaying current songs picture
  - Song picture request
- Displaying and playing a temporary playlist
  - Play the temporary playlist

2. Function Components

After the previous step, it will be more obvious for you how your page is going to look. Next thing is to think about the information that you can display in separate divisions even if they are related. Starting from the App component, up to the grand children, write down the components that you will create. In detailed design of your components, you will update this part to include their children and their children. In addition, when you update your whole app(When adding new functionalities, removing or updating functionalities), you can update this part as required.

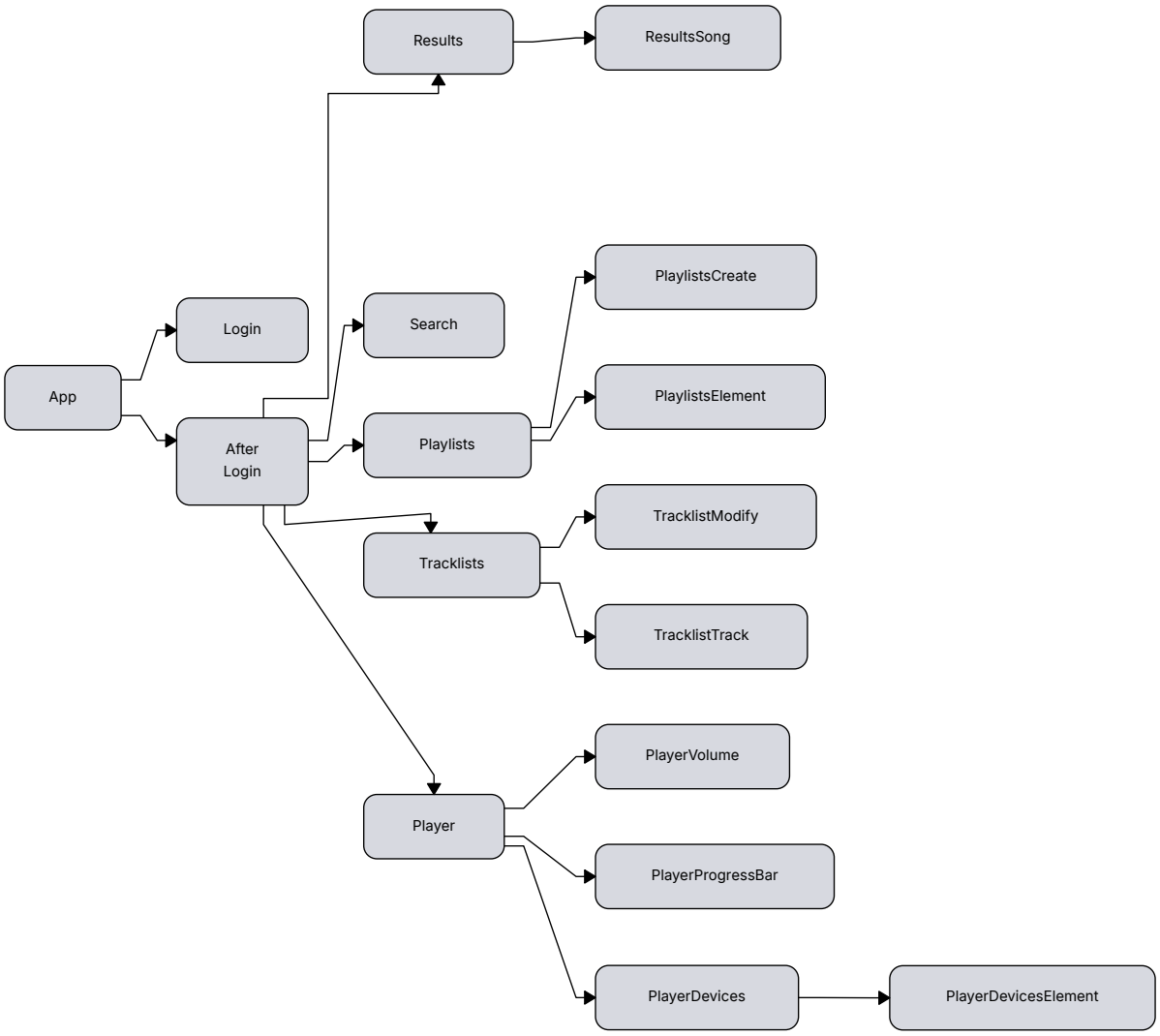


Figure 1: Component Chart

3. Events

Think about all the events that each component will include. Write them down for each component in separate tables. In seperate columns, write down events with identifiable words, event type, event handler names you selected.

Event	Event Type	Event Handler
Sign in	click	handleSignInClick

Table 1: Events in **Login Component and its Children**

Event	Event Type	Event Handler
Search	click	handleSearchClick

Table 2: Events in **Search Component and its Children**

Event	Event Type	Event Handler
Add To Playlist	click	handleAddToPlaylistClick
Remove From Playlist	click	handleRemoveFromPlaylistClick
Resume-Start Song	click	handleResumeOrStartSongClick
Pause Song	click	handlePauseSongClick

Table 3: Events in **Results Component and its Children**

Event	Event Type	Event Handler
Playlist	click	handlePlaylistClick
Create Playlist	click	handleCreatePlaylistClick
Cancel Playlist	click	handleCancelPlaylistClick
Save Playlist	click	handleSavePlaylistClick
Rename Playlist	click	handleRenamePlaylistClick
Cancel Rename	click	handleCancelRenameClick
Apply Rename	click	handleApplyRenameClick
Play playlist	click	handleResumeOrStartPlaylistClick
Pause playlist	click	handlePausePlaylistClick

Table 4: Events in **Playlist Component and its Children**

Event	Event Type	Event Handler
Remove From Tracklist	click	handleRemoveFromTracklistClick
Cancel Edits	click	handleCancelEditsClick
Save Edits	click	handleSaveEditsClick
Resume-start song	click	handleResumeOrStartTrackClick
Pause song	click	handlePauseTrackClick

Table 5: Events in **Tracklist Component and its Children**

Event	Event Type	Event Handler
Skip to Previous	click	handleSkipToPreviousClick
Skip to Next	click	handleSkipToNextClick
Pause	click	handlePauseClick
Start-Resume	click	handleResumeClick
Volume	mouseUp	handleVolumeChangeMouseUp
Repeat	click	handleRepeatClick
Shuffle	click	handleShuffleClick
Show devices	click	handleShowDevicesClick
Select device	click	handleSelectDeviceClick
Close devices menu	click	handleCloseDevicesMenuClick
Reload Devices menu	click	handleRelaodDevicesClick
New		

Table 6: Events in **Player Component and its Children**

4. Detailed Design

Do the detailed design for each component. Detailed design is done on child and grand child components of the App component.

During the design,first, functionalities and fetching opeartions related to this component are first rewritten.

Then the jsx that will be returned is formed. If there is a repetitive part or a part that you find much proper to seperate, for each of these parts, a new child function component is designed. Everything these components will include in terms of jsx are specified here, not in a seperate detailed design section for each child.

After these steps, coding is started, and respective coding operations are executed. During coding, states, effects, and helper functions that will be required are designed(Generally at the beginning of the coding). These must be documented to keep the track of them. This documentation is done on the last page of detailed design, including all of the components in one place. In this page, there will be 3 column grid.

The first column of the grid is for states. In this column there will be 2 column table, 1 column will be for state name, other column will be for where this state is used. In the second column of this table, you need to make the greatest parent bold to place the state in a proper parent.

The second column of the grid is for helper functions. In this column, each component, will have its name as level 3 heading. Under this heading, helper functions will be stated. When stating helper functions, it is necessary to create a different *helperFunction.js* for each current element or its children and their children. Below each file, functions exported should be stated.

The third column of grid is for effect hooks used. Similar to helper functions section, each component's name is used to create level 3 headings. Under these headings, in numbered lists, each effect hook is presented. Each list item includes bullet list with 3 elements. Rows in order are, dependency array, states and state setters and helper functions. All these elements are written when they are used inside these effects.

4.1. App

4.1.1. Functionalities

No functionality

4.1.2. JSX

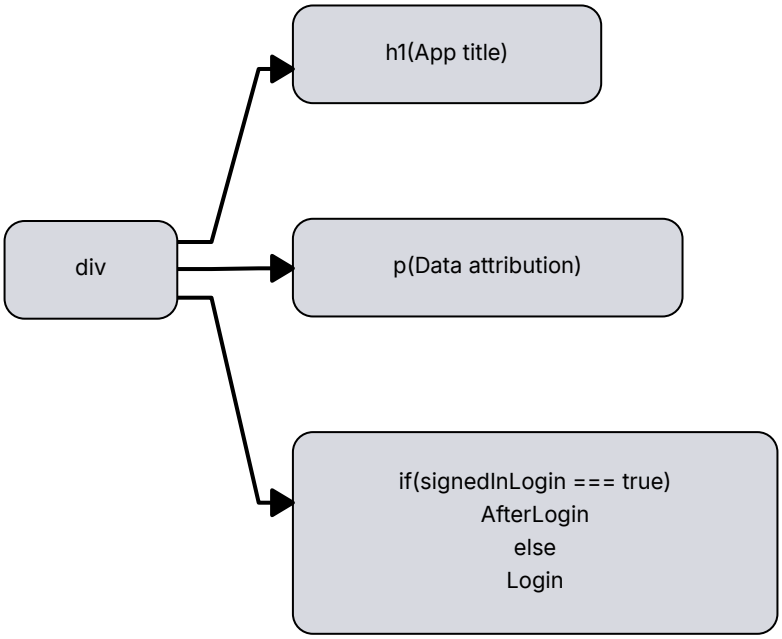


Figure 2: App jsx

4.2. Login

4.2.1. Functionalities

- Sign in click
  - Request token

4.2.2. JSX

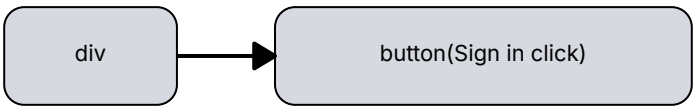


Figure 3: Login jsx

4.3. After Login

4.3.1. Functionalities

No functionality

4.3.2. JSX

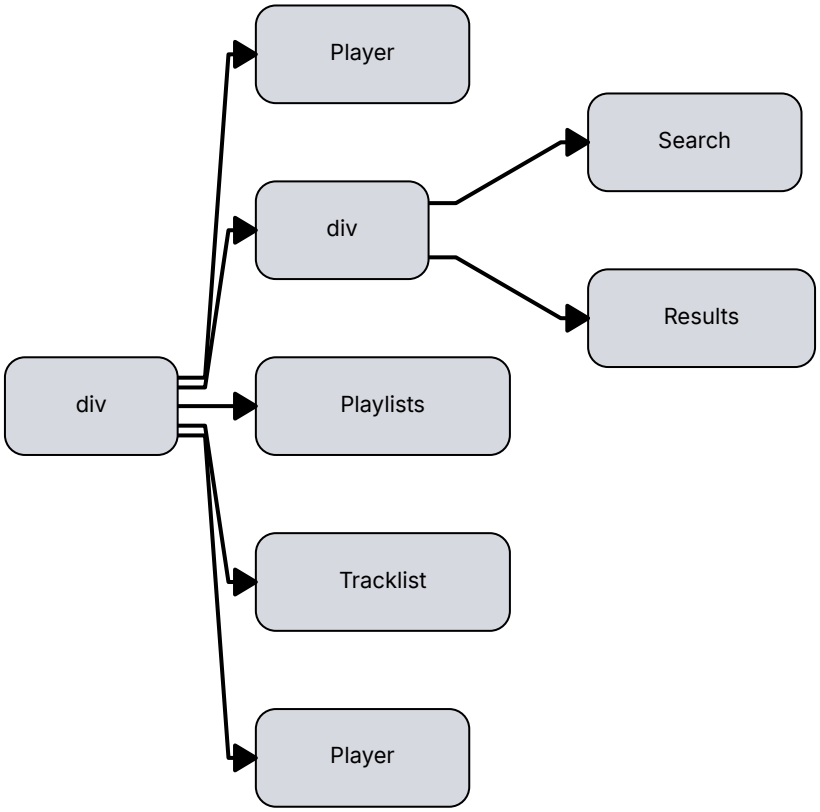


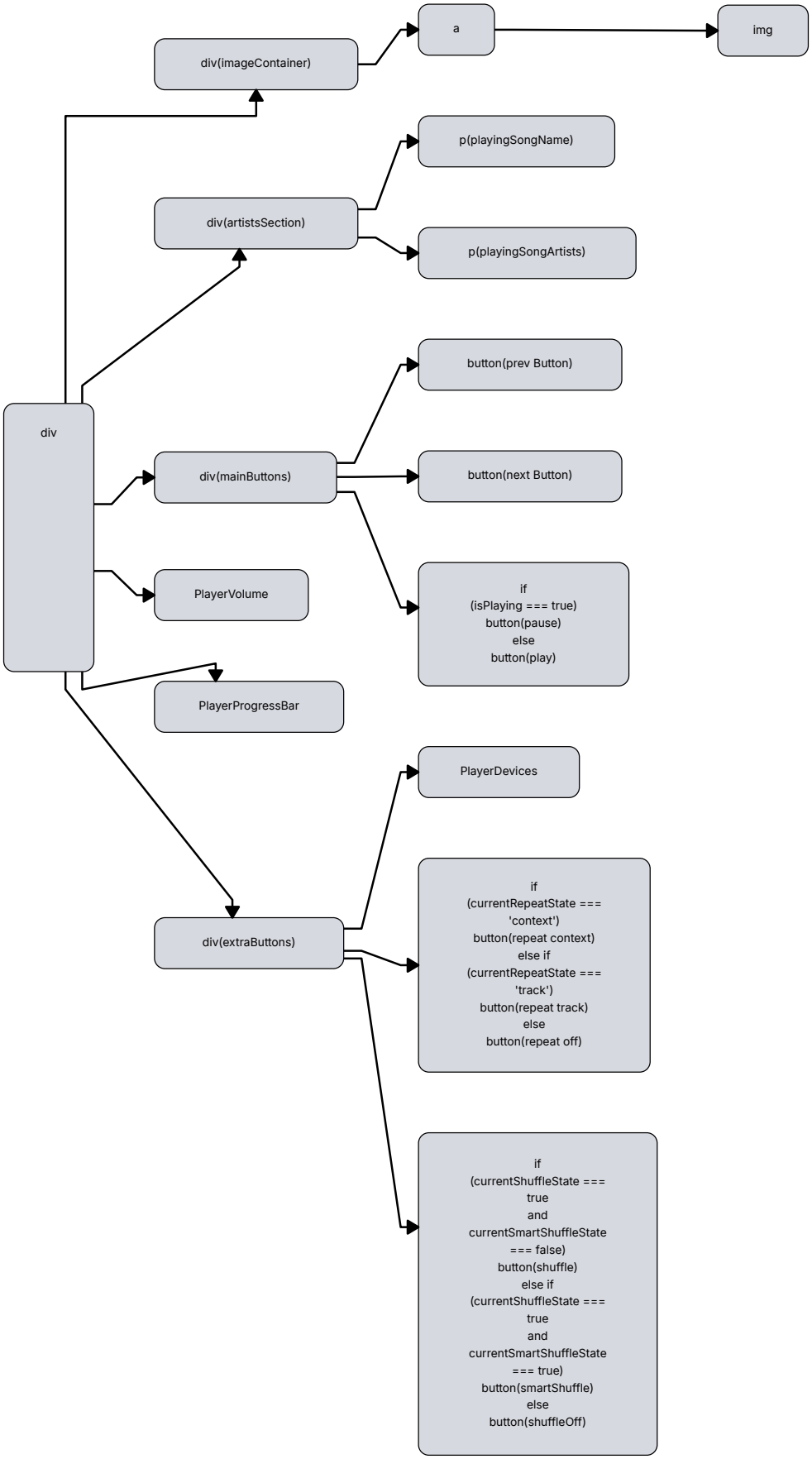
Figure 4: AfterLogin jsx

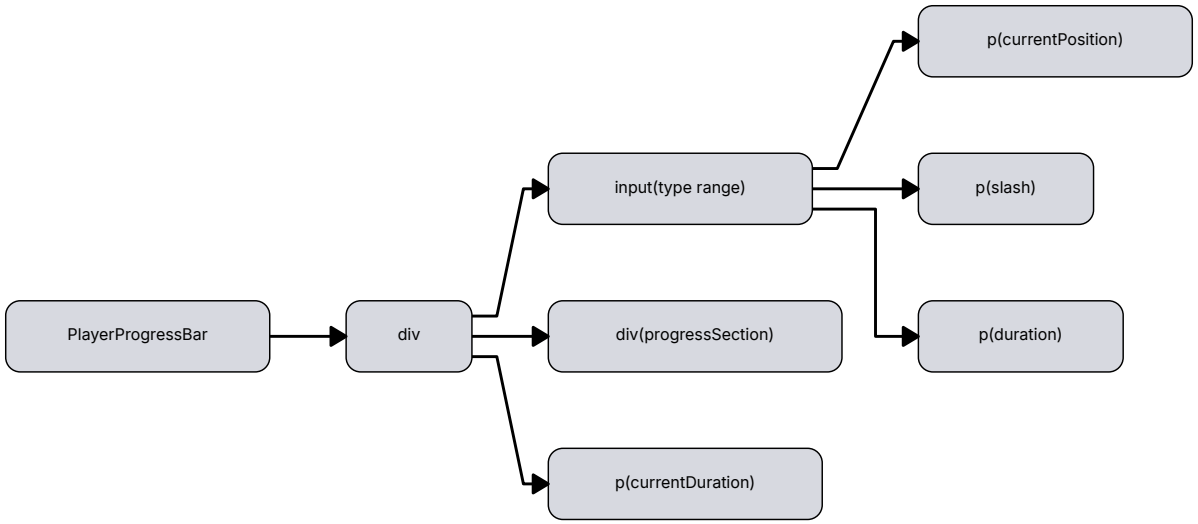
4.4. Player

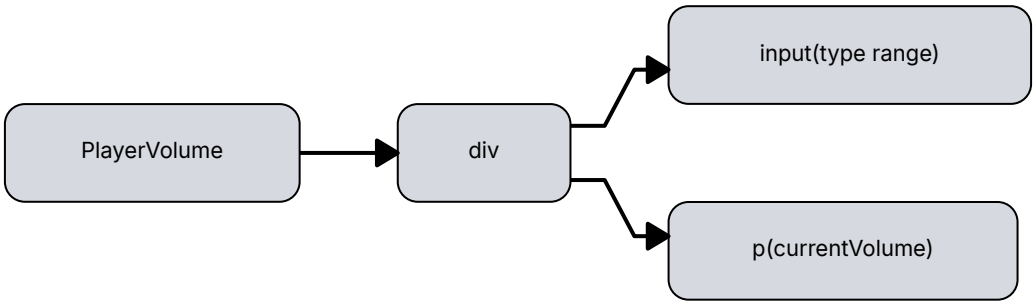
4.4.1. Functionalities

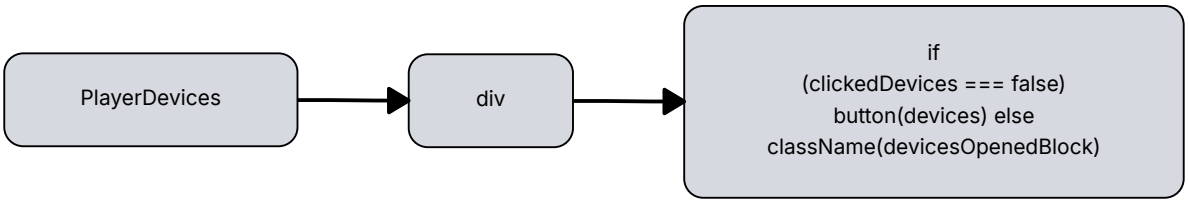
- Skip to previous click
  - Request previous song
- Skip to next click
  - Request next song
- Pause click
  - Request pause song
- Start-Resume click
  - Request start song at desired location
- Volume change
  - Request volume change
- Repeat state change
  - Request repeat state change
- Shuffle state change
  - Request shuffle state change
- Show devices click
  - Request available devices
- Close devices click
- Select device click
  - Request changing current device

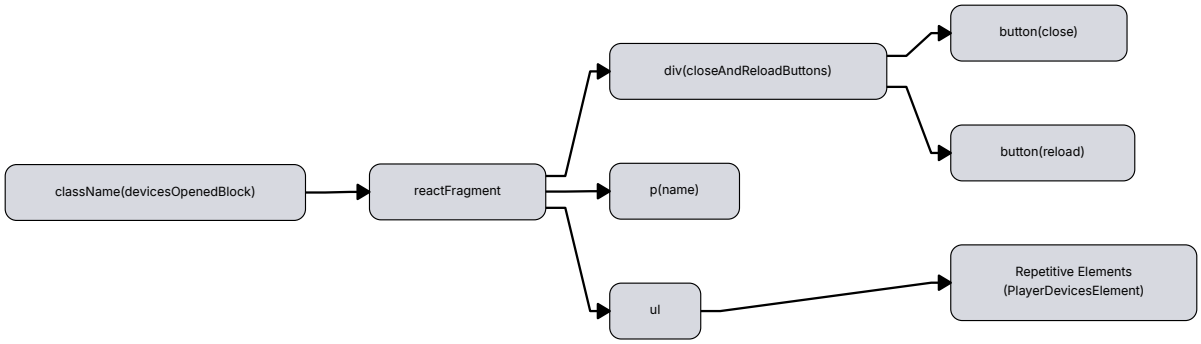
4.4.2. JSX


div
div(imageContainer)
a
img
div(artistsSection)
p(playingSongName)
p(playingSongArtists)
div(mainButtons)
button(prev Button)
button(next Button)
if
(isPlaying === true)
button(pause)
else
button(play)
PlayerVolume
PlayerProgressBar
div(extraButtons)
PlayerDevices
if
(currentRepeatState ===
'context')
button(repeat context)
else if
(currentRepeatState ===
'track')
button(repeat track)
else
button(repeat off)
if
(currentShuffleState ===
true
and
currentSmartShuffleState
=== false)
button(shuffle)
else if
(currentShuffleState ===
true
and
currentSmartShuffleState
=== true)
button(smartShuffle)
else
button(shuffleOff)


PlayerProgressBar
div
input(type range)
p(currentPosition)
p(slash)
div(progressSection)
p(duration)
p(currentDuration)


PlayerVolume
div
input(type range)
p(currentVolume)


PlayerDevices
div
if
(clickedDevices === false)
button(devices) else
className(devicesOpenedBlock)


className(devicesOpenedBlock)
reactFragment
div(closeAndReloadButtons)
button(close)
button(reload)
p(name)
ul
Repetitive Elements
(PlayerDevicesElement)

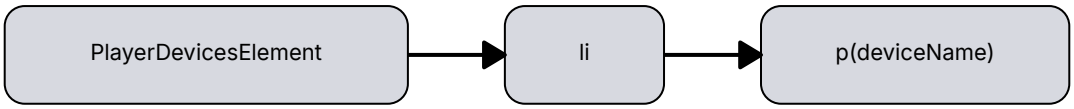


Figure 5: Player jsx

4.5. Search

4.5.1. Functionalities

- Search Click
  - Request search results

4.5.2. JSX

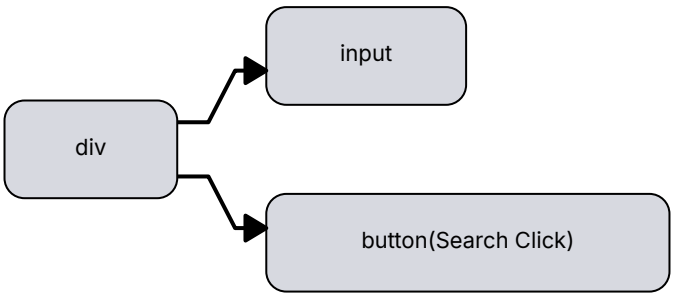


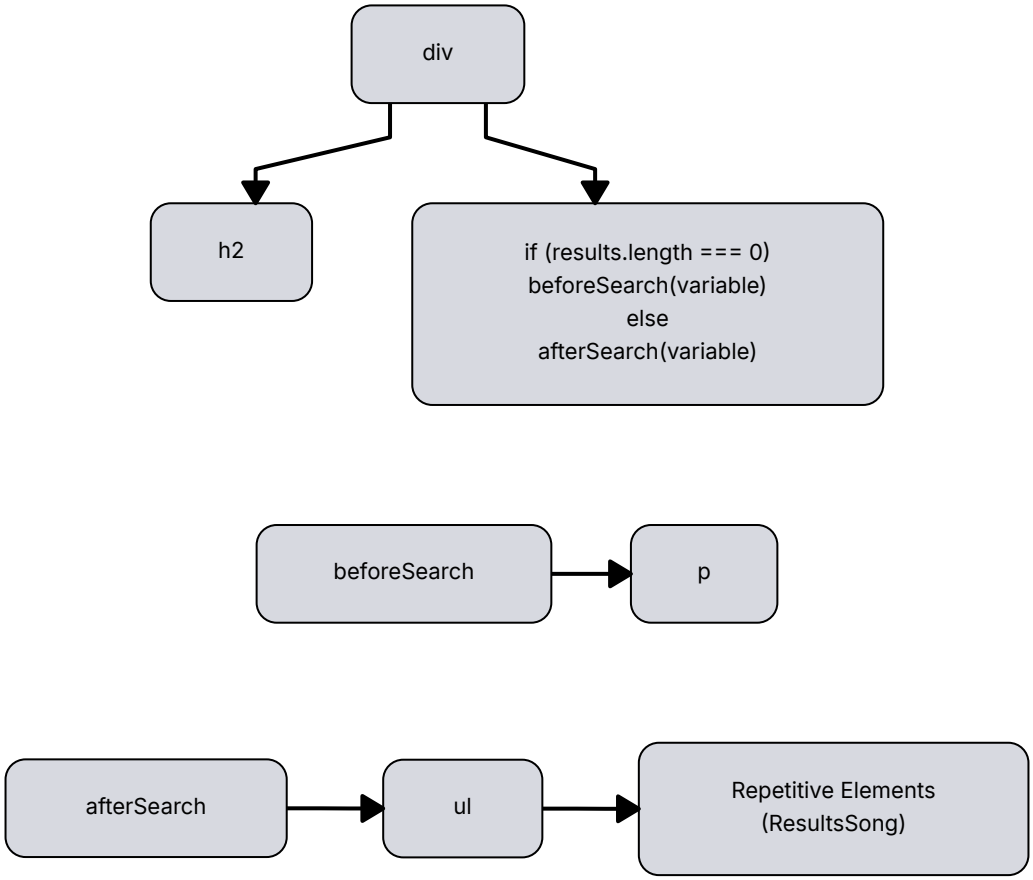
Figure 6: Search jsx

4.6. Results

4.6.1. Functionalities

- Add song click
- Remove song click
- Start-Resume song click
  - Request resuming the song
  - Request new song
- Pause song click
  - Request pausing the song

4.6.2. JSX



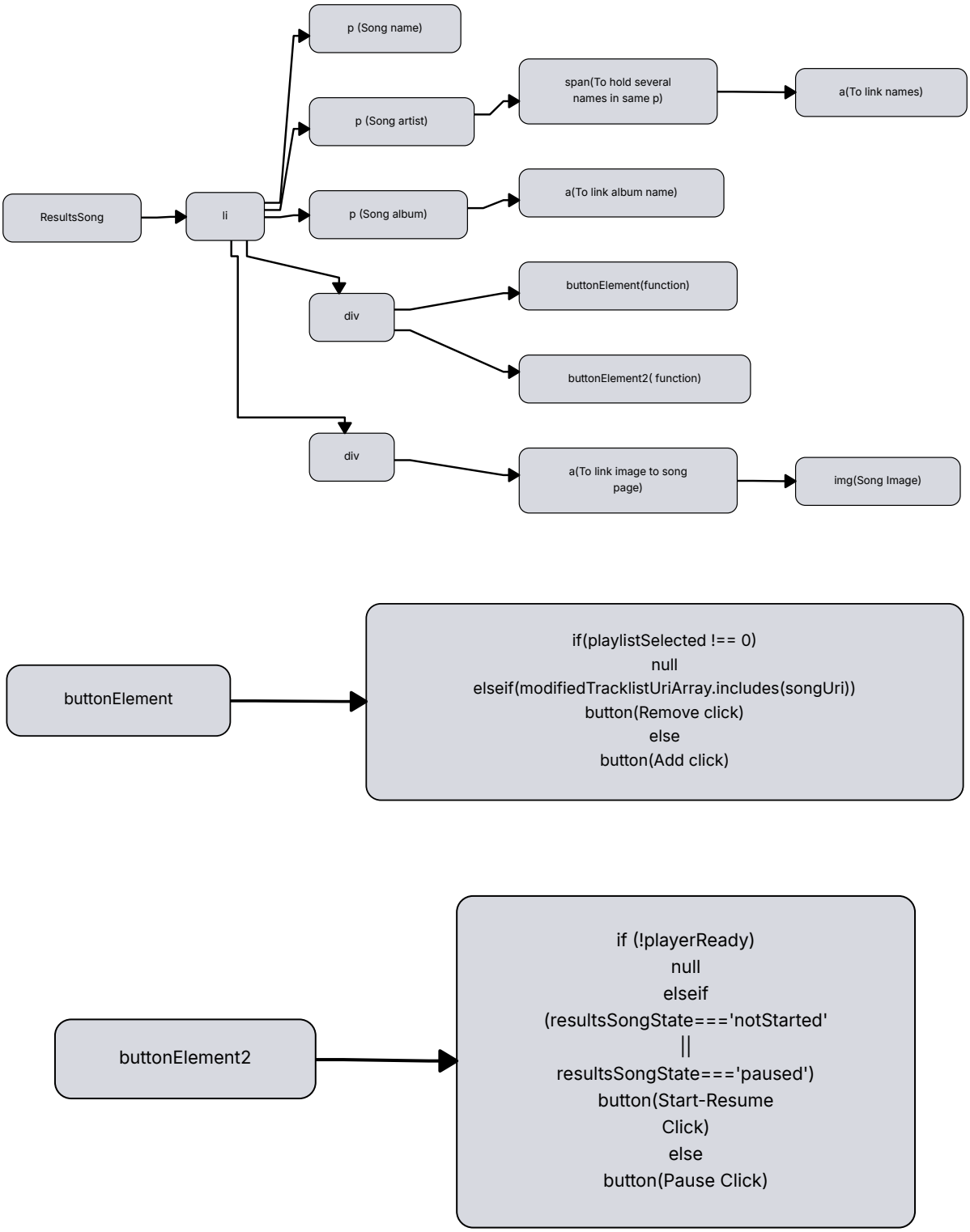


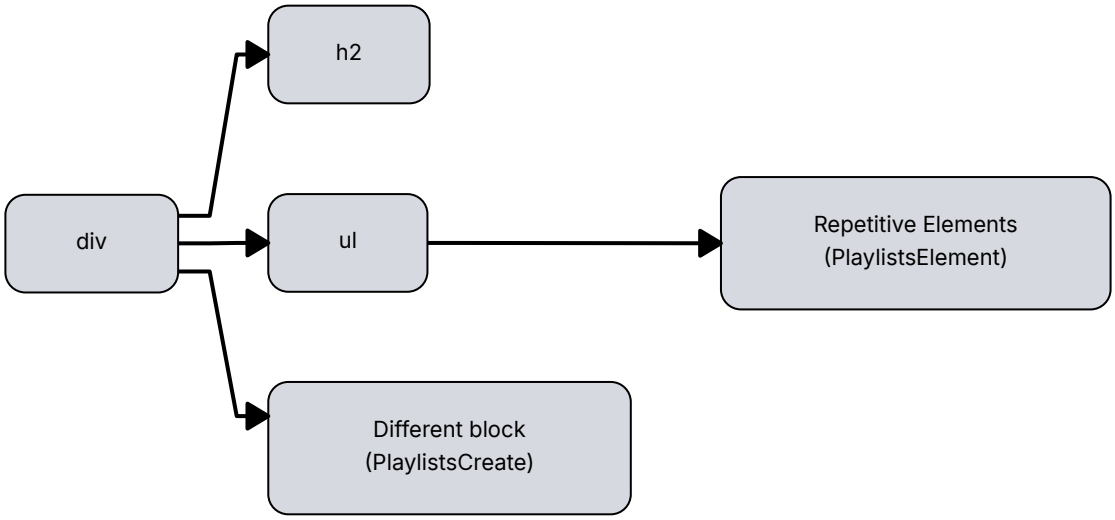
Figure 7: Results jsx

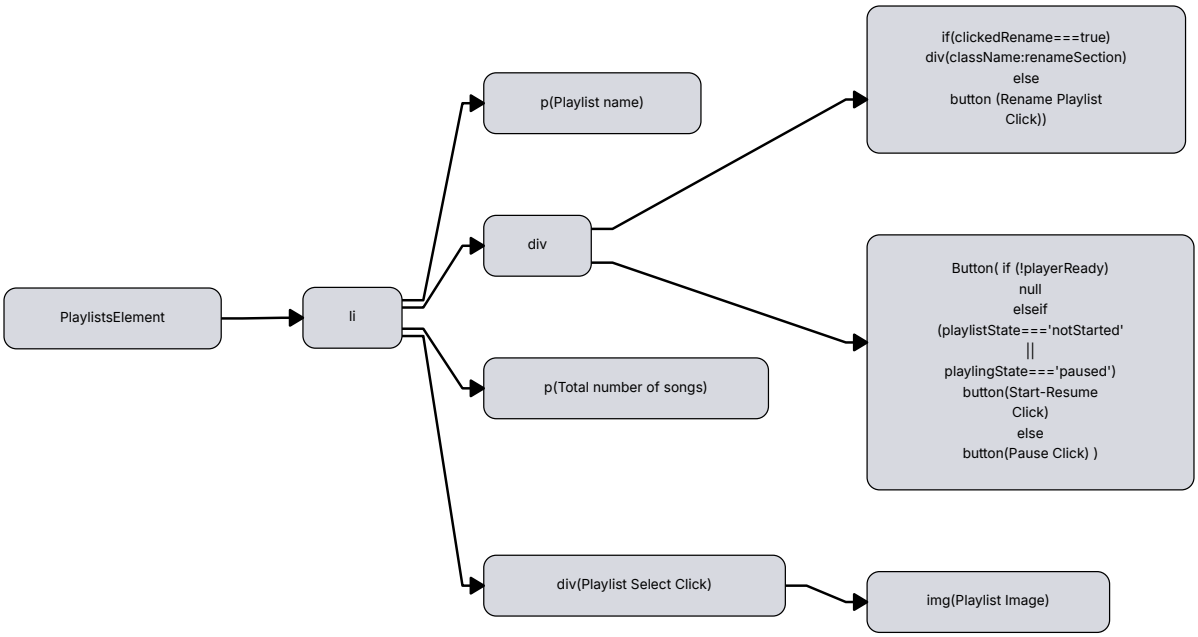
4.7. Playlists

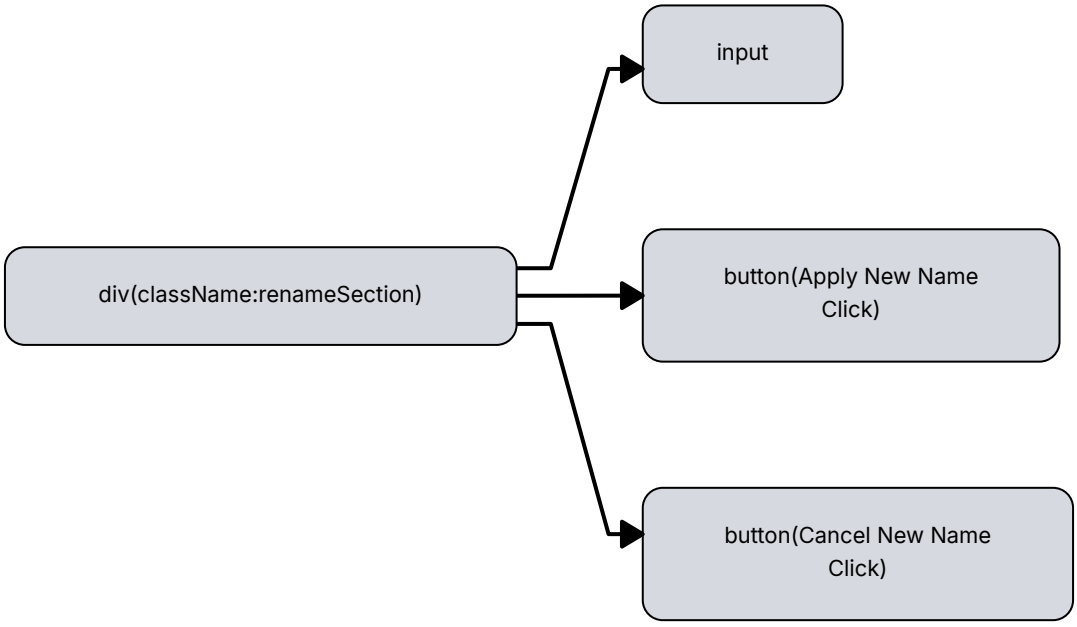
4.7.1. Functionalities

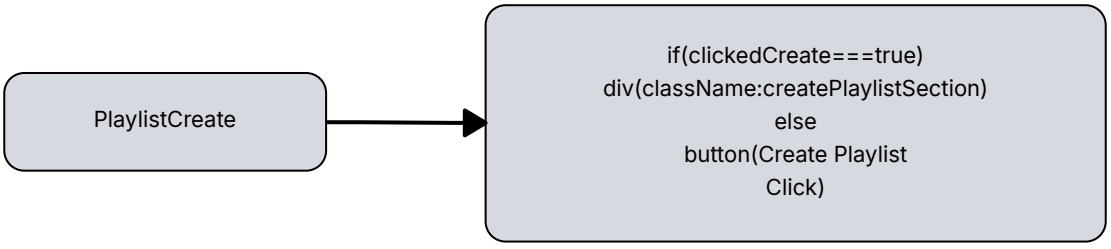
- Request Playlists
- Rename Playlist Click
  - Apply Click
    - Request Playlist Rename
  - Cancel Click
- Create Playlist Click
  - Apply Click
    - Request Playlists
  - Cancel Click
- Select Playlist Click
- Start-Resume Playlist Click
  - Request Playing Playlist
  - Request Resuming Playlist
- Pause Playlist Click
  - Request Pausing Playlist

4.7.2. JSX




PlaylistsElement
li
p(Playlist name)
div
p(Total number of songs)
div(Playlist Select Click)
if(clickedRename===true)
div(className:renameSection)
else
button (Rename Playlist
Click))
Button(if (!playerReady)
null
elseif
(playlistState==='notStarted'
||
playlingState==='paused')
button(Start-Resume
Click)
else
button(Pause Click))
img(Playlist Image)


div(className:renameSection)
input
button(Apply New Name
Click)
button(Cancel New Name
Click)


PlaylistCreate
if(clickedCreate===true)
div(className:createPlaylistSection)
else
button(Create Playlist
Click)

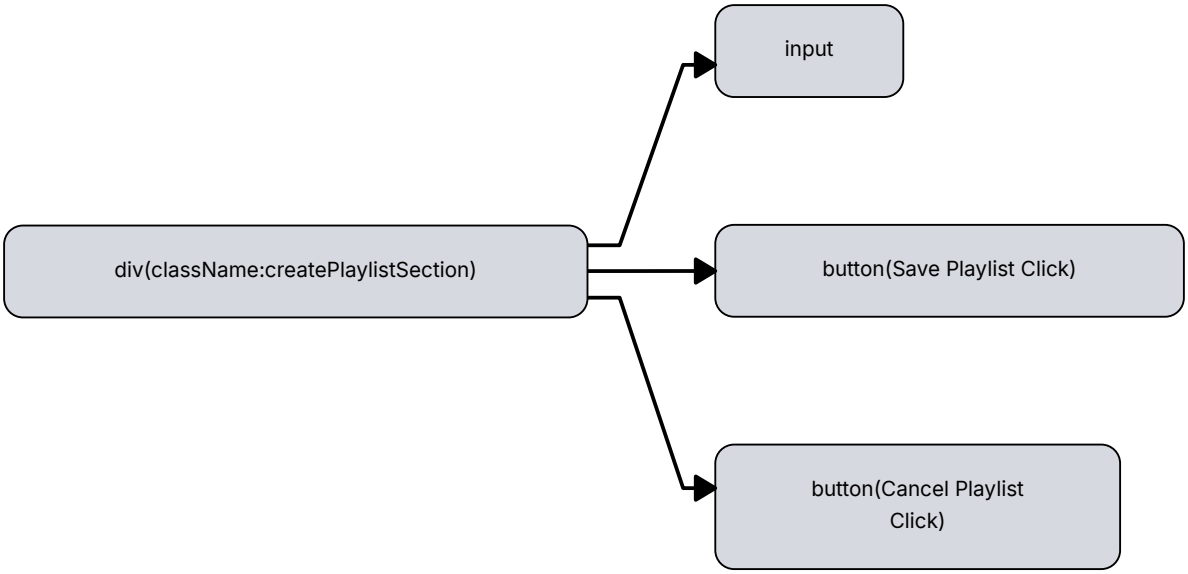


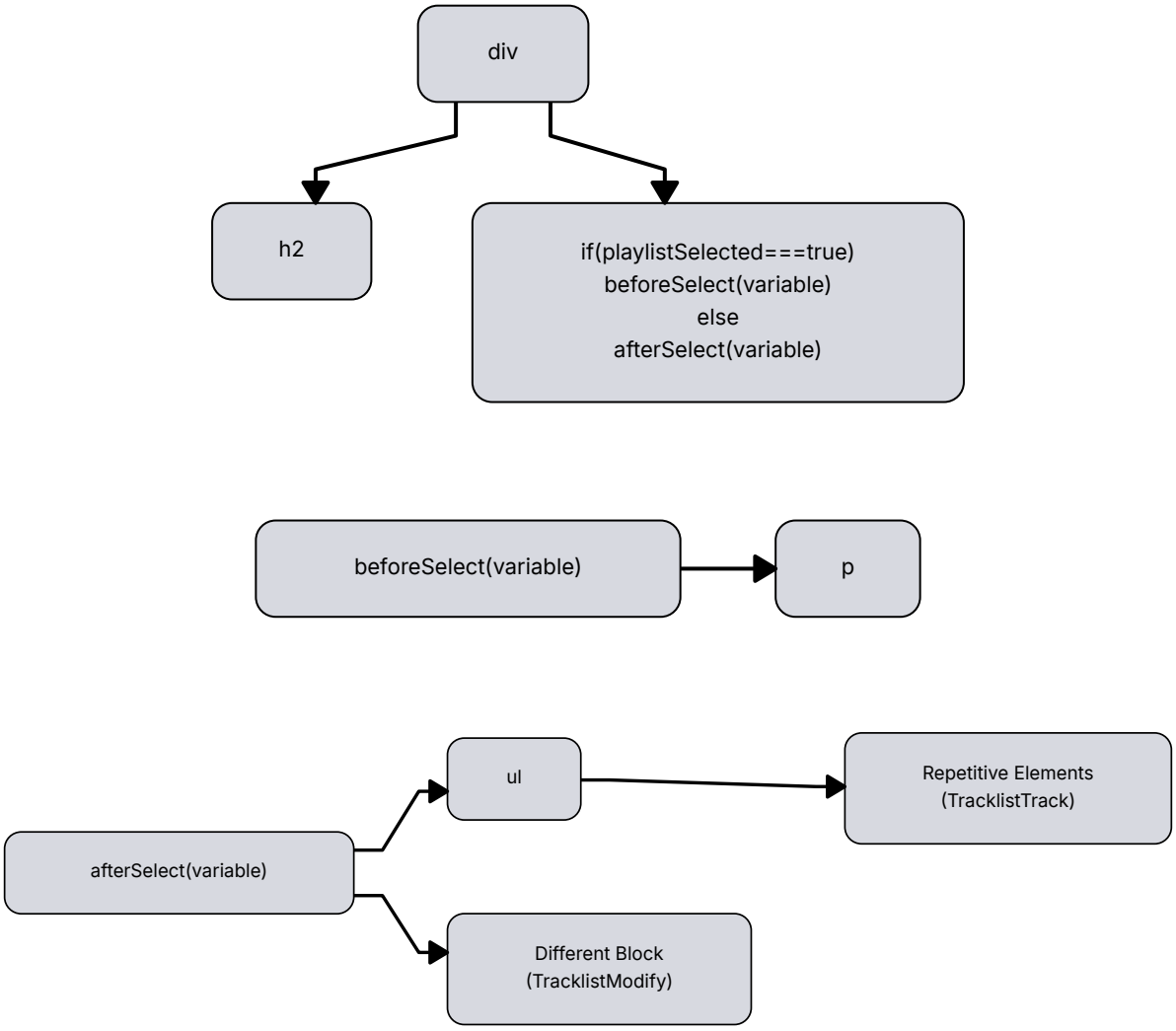
Figure 8: Playlists jsx

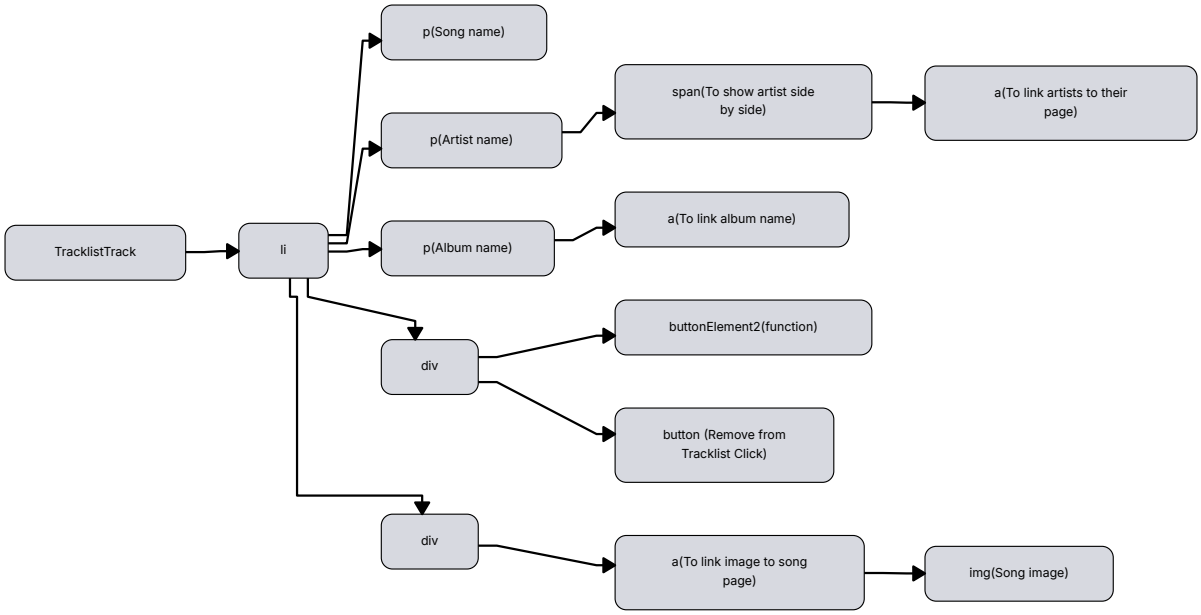
4.8. Tracklist

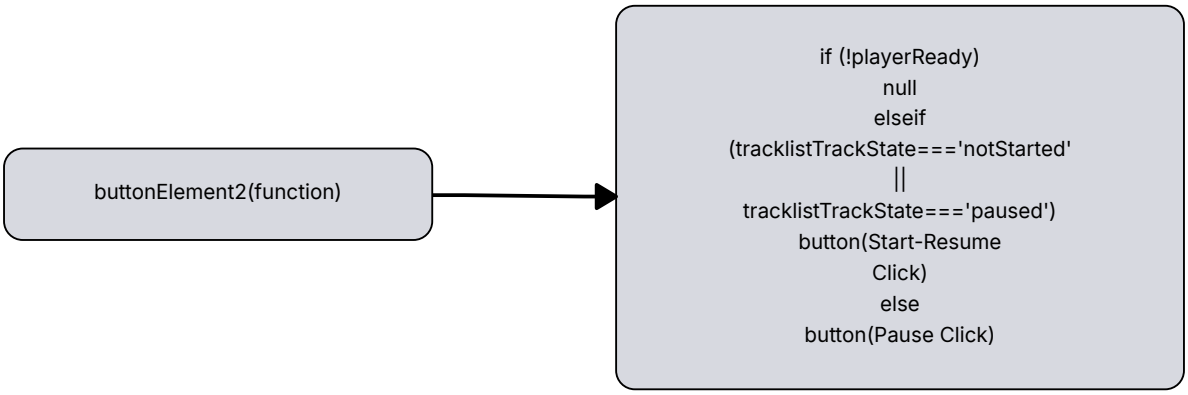
4.8.1. Functionalities

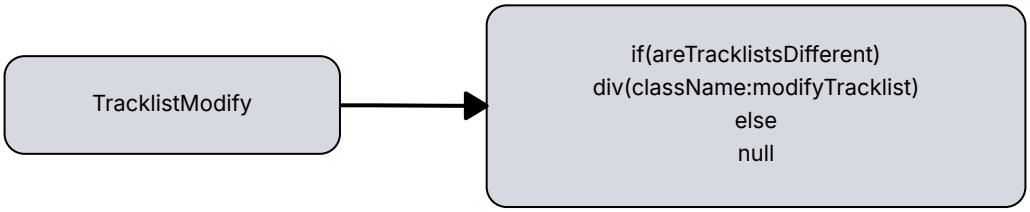
- Request Tracklist
- Remove song click
- Cancel tracklist update click
- Apply tracklist update click
  - Request add song to tracklist
  - Request remove song from tracklist
  - Request Tracklist
- Start-Resume Track Click
  - Request resuming track
  - Request new track
- Pause Track Click
  - Request pausing track

4.8.2. JSX




TracklistTrack
li
p(Song name)
p(Artist name)
p(Album name)
span(To show artist side by side)
a(To link artists to their page)
a(To link album name)
div
buttonElement2(function)
button (Remove from Tracklist Click)
div
a(To link image to song page)
img(Song image)


buttonElement2(function)
if (!playerReady)
null
elseif
(tracklistTrackState==='notStarted'
||
tracklistTrackState==='paused')
button(Start-Resume
Click)
else
button(Pause Click)


TracklistModify
if(areTracklistsDifferent)
div(className:modifyTracklist)
else
null

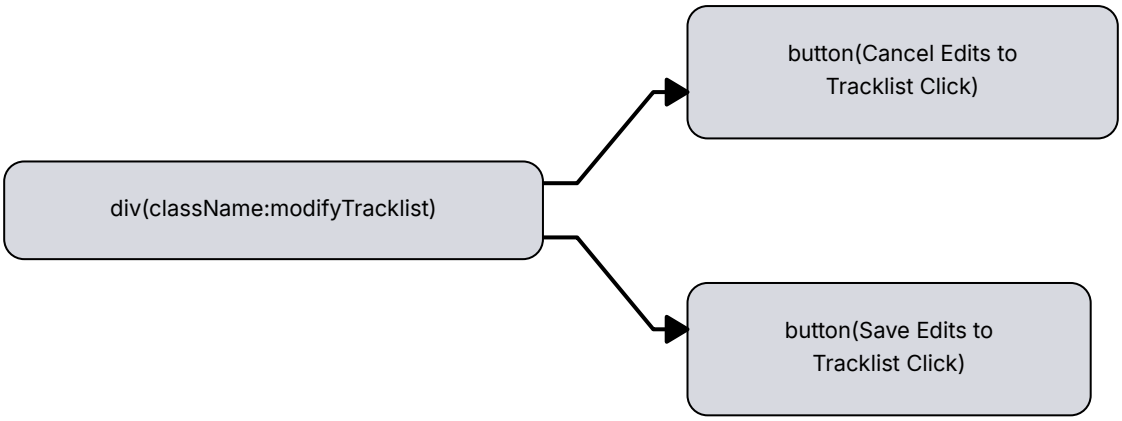


Figure 9: Tracklist jsx

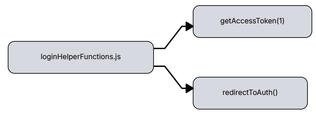
4.9. States

State Name	Function Components
token	Login, **App**, AfterLogin,Search, Playlists, ResultsSong, PlaylistsElement, TracklistTrack, Player, PlayerDevices, PlayerDevicesElement, PlayerVolume, PlayerProgressBar
signedInjamming	Login,**App**
searchString	**Search**
results	Search, Results, **AfterLogin**
modifiedTracklist	Results, **AfterLogin**
playlistSelected	Results, **AfterLogin**, tracklist
clickedRename	**PlaylistsElement**
newName	**PlaylistsElement**
clickNumberTo... TriggerEffect	PlaylistsElement, **Playlists**, PlaylistsCreate
newPlaylistName	**PlaylistsCreate**
clickedCreate	**PlaylistsCreate**
selectedPlaylist	**AfterLogin**, PlaylistsElement, Tracklist, TracklistTrack
originalTracklist	**Tracklist**, TracklistTrack
originalTracklist... UpdateCount	**Tracklist**, TracklistModify
playerReady	ResultsSong, Player, **After Login**, TracklistTrack
playingPlaylistUri	ResultsSong,Player, **After Login**, playlistsElement, TracklistTrack, PlayerProgressBar
playingSongUri	ResultsSong,Player, **After Login**, TracklistTrack, PlayerProgressBar
playingSongPaused	ResultsSong, Player, **After Login**, PlaylistsElement, TracklistTrack
resultsSongState	**ResultsSong**
resultsTitleState	**Results**
currentPosition	ResultsSong, Player, **After Login**, PlaylistsElement, TracklistTrack, PlayerProgressBar
playlistState	**PlaylistsElement**
playlistsTitleState	**Playlists**
tracklistTrackState	**TracklistTrack**
tracklistTitleState	**Tracklist**
playingSongName	Player, **After Login**
playingSongDuration	Player, **AfterLogin**, PlayerProgressBar
currentRepeatState	Player, **AfterLogin**
currentShuffleState	Player, **AfterLogin**
currentVolume	Player, **AfterLogin**, PlayerVolume
updateTriggerCount	Player, **AfterLogin**, ResultsSong, PlaylistElement, TracklistTrack
blocking	**AfterLogin**, ResultsSong, PlaylistsElement, PlaylistCreate, TracklistModify, PlayerDevicesElement, PlayerDevices, PlayerVolume, PlayerProgressBar
playingSongImage	Player, **AfterLogin**
isPlaying	Player, **AfterLogin**, PlayerProgressBar
playingSongArtists	Player, **AfterLogin**
currentDeviceId	PlayerDevicesElement, **AfterLogin**
currentDeviceName	PlayerDevices, **AfterLogin**
availableDevices	**PlayerDevices**
showDevicesClicked	**PlayerDevices**
updateTrigger	**PlayerDevices**
shownVolume	**PlayerVolume**
mouseDown	**PlayerVolume**
shownPosition	**PlayerProgressBar**
shownPositionInMinutes	**PlayerProgressBar**
shownDurationInMinutes	**PlayerProgressBar**
mouseDown	**PlayerProgressBar**
songWhenMouseDownHit	**PlayerProgressBar**
playlistWhenMouseDownHit	**PlayerProgressBar**
trigger	**PlayerProgressBar**
trigger	**PlayerVolume**
initialDeviceId	**AfterLogin**
waiterTrigger	**AfterLogin**

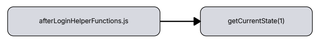
Table 7: States table

4.10. Helper Functions

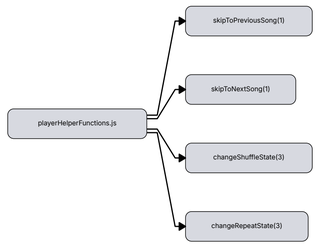
4.10.1. Login



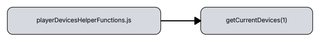
4.10.2. AfterLogin



4.10.3. Player



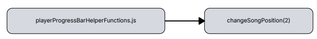
4.10.4. PlayerDevices



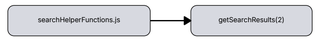
4.10.5. PlayerDevicesElement

4.10.6. PlayerVolume

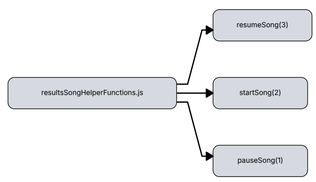
4.10.7. PlayerProgressBar



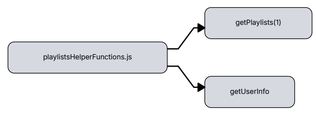
4.10.8. Search



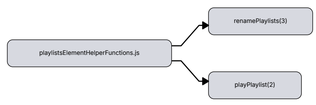
4.10.9. ResultsSong



4.10.10. Playlists

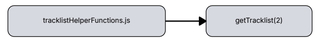


4.10.11. PlaylistsElement



4.10.12. PlaylistsCreate

4.10.13. Tracklist



4.10.14. TracklistTrack

4.10.15. TracklistModify

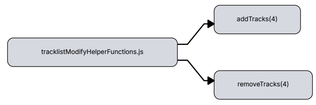


Figure 10: Helper functions

4.11. Effects Section

4.11.1. Login

1.
   - []
   - setToken, setSignedInJamming
   - getAccessToken from (loginHelperFunctions.js),

4.11.2. AfterLogin

1.
   - [updateTriggerCount]
   - setCurretDeviceName, setPlayingSongName, setPlayingPlaylistUri, setPLayingSongUri, setCurrentDeviceId, setPlayingSongPaused, setPlayingSongDuration, setCurrentPosition, setCurrentRepeatState, setCurrentShuffleState, setCurrentVolume,
   - getCurrentState from (afterLogingHelperFunctions.js)
2. • [waiterTrigger]
   - setCurrentDeviceId, initialDeviceId, waiterTrigger
   - None

4.11.3. Player

1. • []
   - None
   - None

4.11.4. PlayerDevices

1.
   - [updateTrigger]
   - setAvailableDevices
   - getCurrentDevices
     from (playerDevicesHelperFunctions.js)

4.11.5. PlayerVolume

1.
   - [currentVolume]
   - setShownVolume, currentVolume
   - None
2.
   - [trigger]
   - setMouseDown, trigger
   - None

4.11.6. PlayerProgressBar

1.
   - [currentDuration, currentPosition]
   - setShownPositionInMinutes, setShownDurationInMinutes, setShownPosition, mouseDown, currentDuration, currentPosition
   - None
2.
   - [mouseDown, shownPosition, isPlaying]
   - setShownPositionInMinutes, setShownPosition, shownPosition, mouseDown, isPlaying
   - None
3.
   - [trigger]
   - setMouseDown, trigger
   - None

4.11.7. Results

1.
   - [results, playingPlaylistUri, playingSongUri, playingSongPaused]
   - setResultsTitleState, results, playingPlaylistUri, playingSongUri, playingSongPaused
   - None

4.11.8. ResultsSong

1.
   - [playingPlaylistUri, playingSongUri, playingSongPaused]
   - setResultsSongState, playingPlaylistUri, playingSongUri, playingSongPaused
   - None

4.11.9. Playlists

1.
   - [clickNumberToTriggerEffect]
   - setPlaylists
   - getPlaylists, getUserInfo from (playlistHelperFunctions.js)
2.
   - [playlists, playingPlaylistUri, playingSongPaused]
   - setPlaylistTitleState,playlists, playingPlaylistUri, playingSongPaused
   - None
3.
   - [clickedCreate, clickedRename]
   - setClickNumberToTriggerEffect ,clickedCreate, clickedRename
   - None

4.11.10. PlaylistsElement

1.
   - [playingPlaylistUri, playingSongPaused]
   - setPlaylistState, playingPlaylistUri, playingSongPaused
   - None

4.11.11. Tracklist

1.
   - [selectedPlaylists, originalTracklistUpdateCount]
   - setOriginalTracklist, setModifiedTracklist
   - getTracklist from (tracklistHelperFunctions.js)
2.
   - [modifiedTracklist, playingingPlaylistUri, playingSongUri, playingSongPaused, selectedPlaylist]
   - setTracklistTitleState, modifiedTracklist, playinging-PlaylistUri, playingSongUri, playingSongPaused, selectedPlaylist
   - None

4.11.12. TracklistTrack

1.
   - [playingSongUri, playingPlaylistUri, playingSongPaused, selectedPlaylist]
   - setTraclistTrackState, playingSongUri, playingPlaylistUri, playingSongPaused
   - None

4.11.13. TracklistModify

1.
   - [modifiedTracklist, areTracklistsDifferent]
   - setOriginalTracklistUpdateCount, modifiedTracklist, areTracklistsDifferent
   - None

5. Seperation of Presentational and Container Components

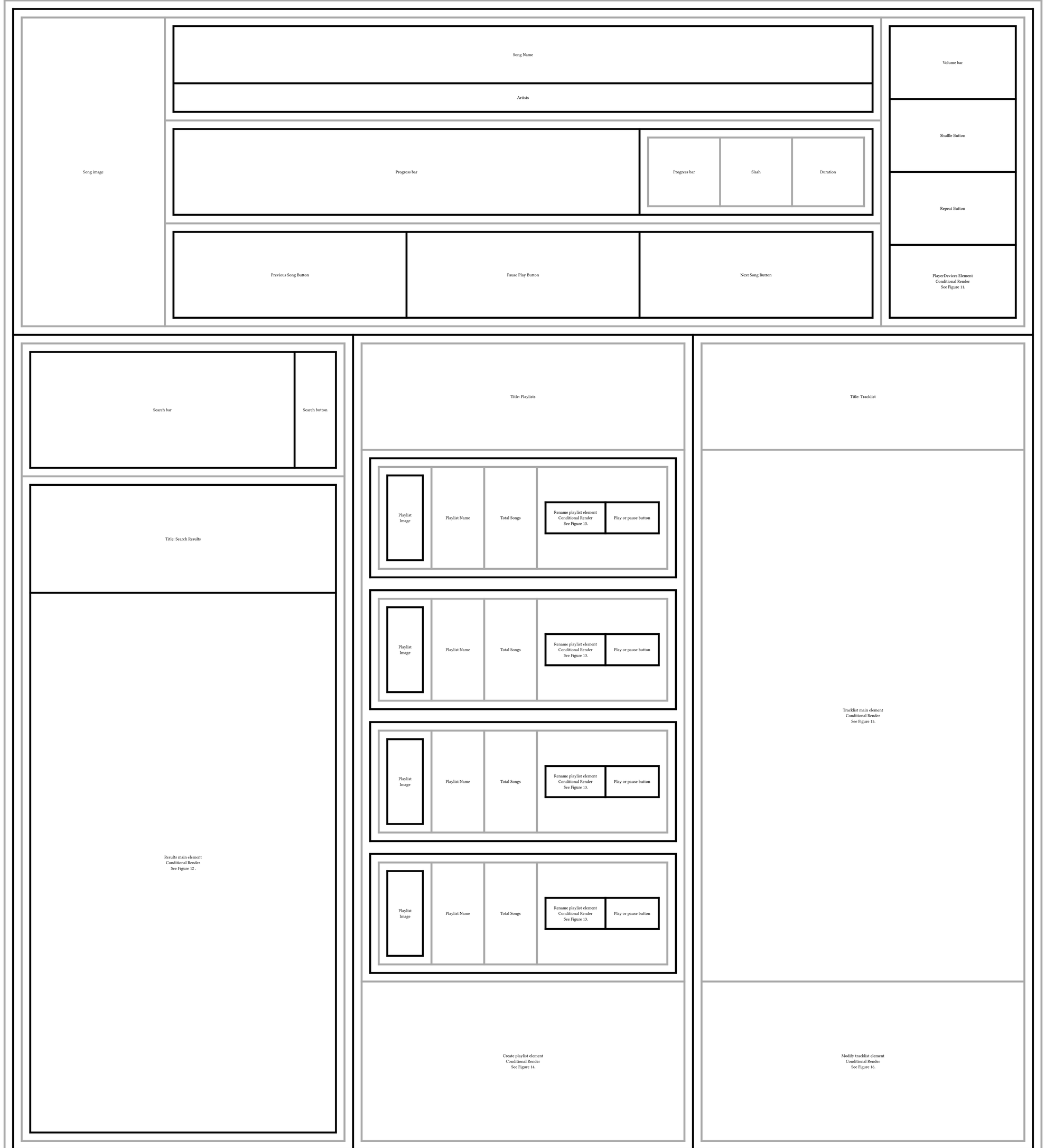
Each react function component should be seperated as presentational and container component.

6. Design of CSS

First, Blueprint for CSS should be detailly designed. Here, the grid elements, and where each button should be placed is determined.

Next, first general css file is created, second the modular css files for each component are determined.

Jammming


Song image
Song Name
Artists
Progress bar
Progress bar
Slash
Duration
Previous Song Button
Pause Play Button
Next Song Button
Volume bar
Shuffle Button
Repeat Button
PlayerDevices Element
Conditional Render
See Figure 11.
Search bar
Search button
Title: Search Results
Results main element
Conditional Render
See Figure 12 .
Title: Playlists
Playlist Image
Playlist Name
Total Songs
Rename playlist element
Conditional Render
See Figure 13.
Play or pause button
Playlist Image
Playlist Name
Total Songs
Rename playlist element
Conditional Render
See Figure 13.
Play or pause button
Playlist Image
Playlist Name
Total Songs
Rename playlist element
Conditional Render
See Figure 13.
Play or pause button
Playlist Image
Playlist Name
Total Songs
Rename playlist element
Conditional Render
See Figure 13.
Play or pause button
Create playlist element
Conditional Render
See Figure 14.
Title: Tracklist
Tracklist main element
Conditional Render
See Figure 15.
Modify tracklist element
Conditional Render
See Figure 16.
Spotify Logo and Attribution

At first, below element is shown.

If any search is done, below element below is shown.

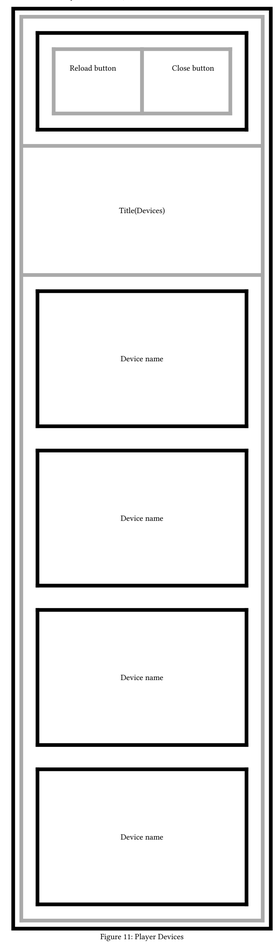


Figure 11: Player Devices

At first, below element is shown.

If any search is done, below element below is shown.

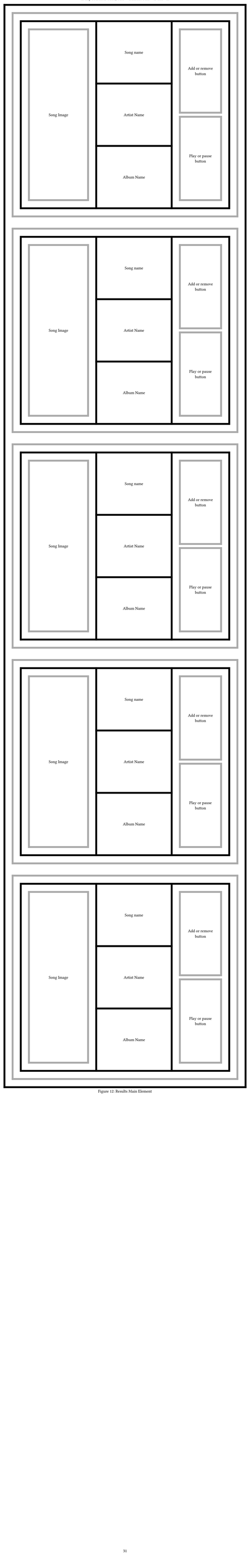
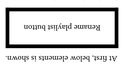


Figure 12: Results Main Element

At first, below elements is shown.



If clicked on this element, element below is shown.

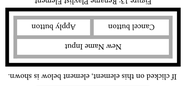
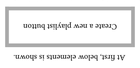


Figure 15: Rename Playlist Element

At first, below elements is shown.



If clicked on this element, element below is shown.

Figure 14: Create PlayList Element

At first, below element is shown.

If any playlist is selected, element below is shown.

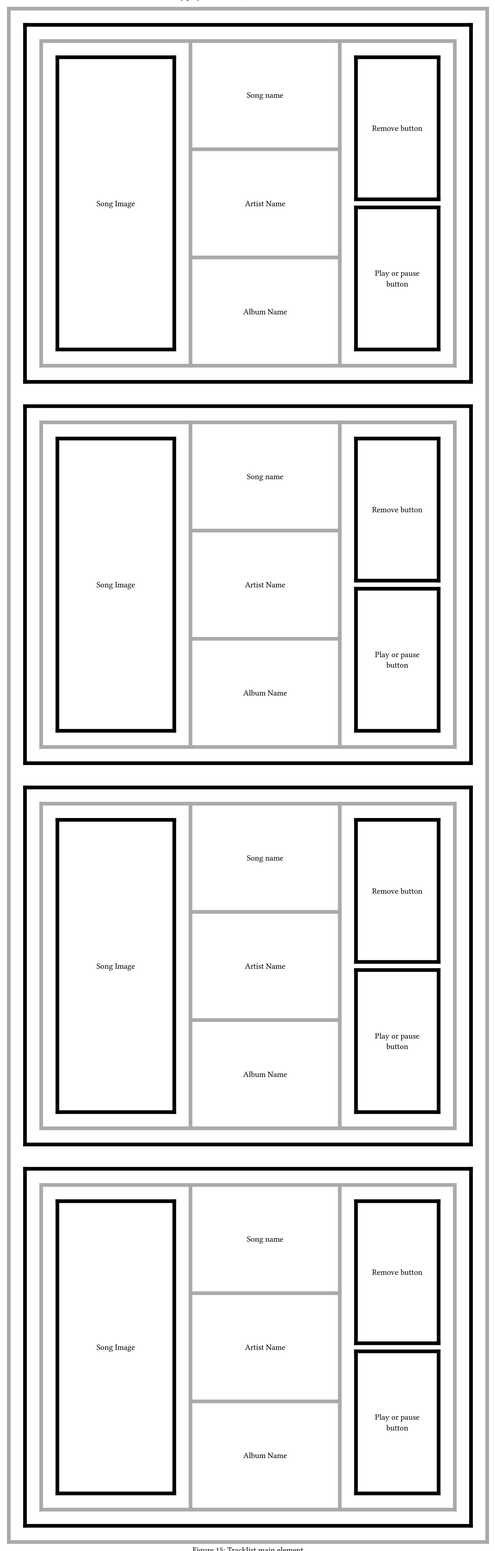


Figure 15: Tracklist main element

At first, nothing is shown.

If any song is added or removed from the tracklist, element below is shown.

Figure 16: Modify Tracklist Element